Long division

1

a $\frac{z}{10} + 1$

b $9x^2 + 5x^4$

c $6x^3 - 4x^2 + 8$

d $8x^3 + 2x^2 - 5x - 7$

2 $\frac{6x^7 - 8x^5}{2x^3} = 3x^4 - 4x^2$

3 $16x^8 - 10x^6 + 12x^4$

4 $x^3 + 0x^2 + 0x + 1$

5

a 28

b $x - 9$

6

a Quotient: $x - 12$, Remainder: 58

b Quotient: $x - 10$, Remainder: 61

c Quotient: $x + 6$, Remainder: 20

d Quotient: $2x^2 + 3x + 4$, Remainder: 0

e Quotient: $3x^2 + 2$, Remainder: 0

f Quotient: $3x + 2$, Remainder: 0

g Quotient: $x^3 - 3x^2 + 9x - 27$, Remainder: 0

h Quotient: $4x^2 + 12x - 5$, Remainder: 0

Synthetic division

7 Yes

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

8

a $x^2 - 4x - 5$

b $x^2 + 3x + 6$

c $x^2 + 5x + 2$

d $3x^2 - 4x + 5 + \dfrac{3}{x - 5}$

e $-3x^2 + 12x + 4 + \dfrac{1}{x + 4}$

f $-3x^2 - 3x + 9 + \dfrac{1}{x + 3}$

g $x^3 + 3x^2 + 4x + 2 + \dfrac{1}{x - 5}$

h $x^3 - 4x^2 - 5x$

i $2x^3 + 3x^2 - 2x$

j $x^3 + 4x^2 - 3x + 2$

k $x^4 - 2x^3 + 4x^2 - 8x + 16$

l $x^6 + 2x^5 + 4x^4 + 8x^3 + 16x^2 + 32x + 64$

9

a $x^2 + 2x + 4$

b $(x + 4)\left(x^2 + 2x + 4\right)$

10

a $x^2 - 3x - 4 + \dfrac{14}{x + 2}$

b 8

11

a

i It represents the remainder when the polynomial P is divided by $x - 5$.

ii $x + 4$

iii 0

b

i It represents the remainder when the polynomial P is divided by $x - 5$.

ii $3x^2 - 3x + 4 + \dfrac{4}{x - 5}$

iii 4

12

a It represents the remainder when the polynomial P is divided by $x - 0.5$.

b $x^2 + 0.5x - 0.75 + \dfrac{1.625}{x - 0.5}$

c 1.625

13

a It represents the remainder when the polynomial $P(x)$ is divided by $x - \sqrt{2}$.

b $x^3 + \sqrt{2}x^2 - 4x - 4\sqrt{2} + \dfrac{3}{x - \sqrt{2}}$

c 3

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

14

a $x - 2$ **b** 0

15

a $x - 4 + \dfrac{1}{x - 5}$ **b** 1

16

a $x^4 + x^3 + x^2 + x + 1$ **b** 0

17 $4\sqrt[3]{3} + 5$

18

a $3x^2 + 2x - 4$ **b** Yes

c $3x^2 + 14x + 100x + \dfrac{816}{x - 8}$ **d** No

19

a $x^3 - 4x^2 + 2x - 3$ **b** Yes

c $x^3 - 12x^2 + 58x - 187x + \dfrac{576}{x + 3}$ **d** No

20

a $x - 4$ **b** Yes

21

a $x - 2 - \dfrac{1}{x + 5}$ **b** No

22

a $3x^2 - 5x + 3 + \dfrac{1}{x + 3}$ **b** No

23

a $3x^2 + 3x$ **b** Yes

24

a $2x^2 + 2x + 2$ **b** Yes

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

25

a $8x^2 + 4x + 4 - \dfrac{1}{x - 0.25}$

b No

26

a $3x^5 + 3\sqrt{2}x^4 + 5x^3 + 5\sqrt{2}x^2 + 12x + 12\sqrt{2}$

b Yes

27

a $\pm 4,\ \pm 1,\ \pm 2,\ \pm \dfrac{1}{2}$

b No

c Yes

d $(2x + 1)\left(x^2 - x - 4\right) = 0$

e $\dfrac{1}{2}, \dfrac{1 \pm \sqrt{17}}{2}$

28

a $\pm 1,\ \pm 2,\ \pm 3,\ \pm 6$

b No

c Yes

d $(x - 1)(x - 2)(x - 3)$

e $x = 1, 2, 3$

29

a $x = 1$

b $6x^2 - 5x + 1$

c Yes

d $x = 1,\ \dfrac{1}{2},\ \dfrac{1}{3}$

30

a $x = 1$

b $x^2 + 4x + 3$

c Yes

d $(x - 1)(x + 1)(x + 3)$

e $x = 1,\ -1,\ -3$

Applications

31

a $\dfrac{0.2x^3 + 0.6x^2 + 0.45x + 0.1}{x + 0.5}$

b $0.2x^2 + 0.5x + 0.2$ units

32 $x^3 - 3x^2 + 2x - 2$ in

33 $\left(x^3 - 2\right)$ units

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

34 $\left(3x^3 + 2x^2 + 5x + 7\right)$ units

35 $\left(x^2 + x - 4\right)$ units

36 $\left(26n^2 + 22n + 58\right)$ units

37 $\left(2x^2 + 6x + 1\right)$ units

38 $\left(2x^2 - 4x + 9\right)$ hours

39 $\left(4x^4 - 3x^3 + 8x^2 - 6x + 7\right)\ \mathrm{m}^2$

40 4 days

© ClipSAT: By Mr. Mohamed Abdallah +966597688647